Research Paper Context

Everything a person or an LLM needs to write our paper and presentation on site-specific smartphone anemia screening, apart from our results.

Project	Do tongue and nail-bed images add anything beyond the conjunctiva for smartphone anemia screening?
Working title	Site-specific deep learning for smartphone anemia screening: re-examining whether tongue and nail-bed images add value beyond the conjunctiva
Base paper	Desai S, et al. Smartphone-based Non-Invasive Classification of Anemia Severity using Deep Learning. medRxiv preprint, 26 Oct 2026 [1]
Dataset	SEWA Rural multi-modal anemia dataset, gated on Hugging Face, CC BY-NC 4.0 [5]
Prepared	1 November 2026, while the first training run is in progress
Code	DS-Research-Implementation repository (remote thanay2007/DS-Research-Implementation); nothing from the current work is committed yet

What this document leaves out on purpose. It has no results from our experiments. Tables and figures for them are listed in Section 8 with blanks to fill from `data/results/` once `evaluate.py` has run. Every result quoted here belongs to the base paper or another cited study and is labelled as such. The only numbers from our own work are the cohort description, crop-detection rates and tuning runs, because the Methods section has to report them.

Read Section 7 before writing Methods or drawing conclusions. While preparing this document we found that the base paper's authors published code [3] that our README does not mention. It shows that our running model differs from theirs, and several other planned steps are not built yet. Text written now must not describe them as done.

If you are an LLM reading this

1. Never invent results, confidence intervals, p-values or patient counts. If a number is not in this document, write **[RESULT PENDING]**.
2. Do not describe a step in Section 7 as completed unless you are told it has since been done.
3. Keep the base paper's numbers separate from ours, and never compare them as if they came from one split. Compare against our own retrained baseline on our folds (Section 6.8).
4. Say "did not improve" only with its confidence interval. Say "redundant" only if the diagnosis test in Section 5.4 supports it.
5. Check every reference against its source before it goes into the paper. Section 10 explains how far each one has been checked.
6. Never ask for, repeat or upload raw patient photos, contact sheets or row-level data. The dataset policy bars redistribution.

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1. The project in one page

The problem. Anemia is common and under-diagnosed in low-resource settings. Testing normally needs a blood sample, equipment and trained staff. A phone camera could screen people without a needle by reading pallor in body sites where blood is visible through thin tissue.

What the base paper did. SEWA Rural and Google collected conjunctiva, tongue and nail-bed photos and fingertip PPG from hospital patients in rural Gujarat, with venous lab hemoglobin (Hb) as ground truth. Their final model uses only the conjunctiva with age, sex and pregnancy status. They concluded that adding tongue and nails "did not improve performance" [1].

Why we are not satisfied with that. The conclusion rests on one appendix table of point estimates, with no confidence intervals or significance tests, one validation split, one fusion method, and only a conjunctiva-specific preprocessing recipe described. "Did not help" has not been shown to mean "redundant".

What we do. We test the claim properly on the released data:

- Train all 7 combinations of the 3 sites with 5-fold, person-level cross-validation, and compare them with confidence intervals and paired tests.
- Run every combination through two preprocessing pipelines: the conjunctiva recipe applied to everything (**shared**) and a crop tailored to each site (**specific**).
- Where a site adds nothing, work out why: no signal, duplicate signal, a model or fusion failure, or bad images.
- Check whether the answer changes for children, pregnant women, severity bands, lighting and phone model.

Key facts

Item	Value
Ground truth	Venous Hb on a Sysmex XN-550 analyser, column `cbc_hgb_g_d1`
Sites and photos	C = conjunctiva (1 photo), T = tongue (1 photo), N = nail bed (2 photos: open and closed hand)
Model output	A continuous Hb value in g/dL, then WHO cut-offs by age, sex and pregnancy give the anemia band
Combinations	C, T, N, C+T, C+N, T+N, C+T+N
Scale	7 combinations x 5 folds x 2 pipelines = 70 runs; 65 are trained because C is identical in both pipelines
Run time	About 25 hours for the shared pipeline and 24 for the specific pipeline on one RTX 3050 laptop GPU
External test	The base paper's field-representative group, held out completely
Out of scope	PPG (about 650 GB, and the base paper found comparable or lower accuracy), so it is future work

Glossary

Term	Meaning
Hb	Hemoglobin concentration in blood, in g/dL. The quantity we estimate.
Conjunctiva	The inner lining of the lower eyelid. Pallor here is a classic clinical sign of anemia.

Term	Meaning
MAE	Mean absolute error: the average size of the Hb error in g/dL. Lower is better.
AUC / AUROC	How well a score ranks positive people above negative people. 0.5 is chance and 1.0 is perfect.
Sensitivity, specificity	The share of anemic people correctly flagged, and the share of non-anemic people correctly cleared.
Bland-Altman	A plot of prediction error against the mean value, with bias and limits of agreement (bias ± 1.96 SD).
CIELAB	A colour space with lightness (L*) and two colour axes (a* green-red, b* blue-yellow), closer to human perception than RGB.
Fold	One of 5 equal groups of people. Each model tests on one fold it never trained on.
Fusion	How features from several images are combined. Concatenation joins them before the final layers, late averaging averages separate predictions, and attention learns a weight for each site.
Shapley value	A site's average marginal contribution across all combinations of the other sites.
HemoCue	A point-of-care Hb test using a drop of blood. Our invasive reference comparison.
PPG	Photoplethysmography: a fingertip-over-camera video signal.

Spelling. The base paper and our README use American "anemia", while WHO and the project folder use "anaemia". This document uses "anemia". Pick one for the paper to match the venue.

2. Background for the Introduction

2.1 Why anemia matters

The figures below are as stated by the base paper, which cites them to its references [7, 8, 9]. Confirm each against the primary source before citing it.

- Anemia affects over two billion people globally.
- Globally, about 40% of children under five, 37% of pregnant women and 30% of women of reproductive age are affected.
- In India, NFHS-5 (2019-21) reports prevalence as high as 67% in children and 57% in women [9].
- Treatment works when anemia is detected, so detection is the bottleneck. Severe anemia is the most time-sensitive group.

2.2 Why current detection falls short

Method	Problem	Source
Lab analyser (venous CBC)	Accurate, but needs phlebotomy, a laboratory and logistics. Hard to reach in remote settings.	[1]
Point-of-care test such as HemoCue	Accurate, but needs a blood sample, skilled staff, supply chains and stable reagents.	[1]
Clinical examination for pallor	Needle-free and cheap, but unreliable. In the base paper's test set it had 73% sensitivity and 53% specificity for any anemia, and only 31% sensitivity for severe anemia.	[1, 10, 11]
Smartphone images	Needle-free and scalable. Earlier studies were limited by small, homogeneous, often unavailable datasets or weak reference standards.	[1, 12]

2.3 What a useful tool would need

Before their study, the base paper's authors asked state programme officers, public health practitioners and non-profits what would be acceptable [1, 28]. They set a target of **at least 80% sensitivity and specificity** for the

anemia categories, especially severe anemia. Predictions should arrive within 30 seconds, preferably offline, on low-cost phones used by a community health worker after minimal training. This target is the natural benchmark for our Discussion.

2.4 Why the site question matters in practice

- **Conjunctiva capture is not always easy.** The base paper reports trouble getting clear, retracted eyelid images from children aged 1-5 because of motion. A fallback site would help them directly.
- **Every extra photo costs field time.** If tongue or nails add nothing, the workflow should drop them. If they add something, dropping them loses accuracy.
- **Nails and tongue have their own failure modes.** These include henna, polish, cold fingers, saliva glare and staining.
- **The published answer may shape future tools and datasets.** A weak "did not help" could close off a line of research too early.

3. The base paper in detail

3.1 Citation and links

Desai S, Gadgil S, Gottala VA, Jesalpura K, Ramnath D, Gruzewska E, Purandare H, Liu Y, Telang U, Taylor JA, Sawhney R, Tiwari R, Webster DR, Shetty S, Hassidim A, Matias Y, Howell M, Venkatraman S, Daryani B, Chen C, Kakarmath S, Daswani M, Wilson M, Pullakhandam R, Vispute S, Thompson M. *Smartphone-based Non-Invasive Classification of Anemia Severity using Deep Learning: A Cross-Sectional Study in Rural India.* medRxiv, posted 26 October 2026. It is a preprint and has not been peer reviewed. DOI: 10.64898/2026.10.22.26366247.

3.2 Study design

Aspect	What the paper reports
Design	Prospective cross-sectional study at SEWA Rural hospital, serving more than 2,500 villages in rural indigenous Gujarat. Funded by the Gates Foundation (INV-043120).
Ethics	SEWA Rural Institutional Ethics Committee approval on 31 March 2023, CTRI/2023/07/055313, and Health Ministry Screening Committee clearance. Written informed consent.
Participants	Ages 1-89, including pregnant women, with a clinician-ordered CBC during routine care. Non-consecutive sampling, enriched for severe anemia, with a target of about 25% in each of four Hb categories. People with non-removable nail material such as henna were not enrolled.
Numbers	5,455 captures from July 2024 to November 2025. 279 excluded for incomplete metadata, images or crop errors, leaving 5,176. Split 70:15:15 into 3,623 train, 776 validation and 777 test. Field-representative group: 516 images from 429 people, December 2025 to January 2026.
Composition	Normal 27.0%, mild 25.2%, moderate 34.1%, severe 13.7%. 83.7% female. Mostly outpatients from obstetrics and gynecology (56.8%) and general medicine (31.5%).
Phones	Samsung A05, A14 and S24, covering low to high cost.
Capture	Flash always on. A white paper reference above the eye or under the hand. One eye with lower lid gently pulled, protruding tongue, and nails with the hand flat (open) and with fingers flexed (closed). Removable polish cleaned off. Also a 30-second fingertip PPG.
Reference standards	Ground truth: venous Hb on Sysmex XN-550. Comparators: HemoCue Hb 301 on venous and capillary blood, and staff clinical examination (staff were not blinded to Hb).
Quality control	A pathologist reviewed images. Low-quality or identity-revealing images were removed.
Other measures	Skin tone by spectrophotometer as ITA°, lux meter readings, demographics, symptoms and vitals.

3.3 Model and training as published

Input images are cropped under the white paper edge using Canny edge detection, resized to 256 x 256 and converted to CIELAB. A two-stage ResNet with one block per stage, trained from random initialisation, produces features that are concatenated with age, sex and pregnancy status. An MLP outputs continuous Hb. Training uses Adam with exponential decay, Huber loss and inverse-bin-frequency reweighting, plus one constant bias correction chosen on validation. Hb is then thresholded at WHO cut-offs (Appendix 4). Section 6.9 compares every detail of this with our implementation.

3.4 Published results on the conjunctiva model

Measure (test set, n = 777)	AI model	HemoCue	Clinician exam
Hb MAE, g/dL	1.12 (normal 1.19, mild 0.74, moderate 1.23, severe 1.43)	Venous 0.48, capillary 0.52	n/a
Within ±1.0 g/dL	54%	not stated	n/a
Bias, limits of agreement	-0.02; -2.90 to +2.85 g/dL	not stated	n/a

Measure (test set, n = 777)	AI model	HemoCue	Clinician exam
Any anemia: sensitivity, specificity, AUC	70%, 70%, AUC 0.80	Venous 86%, 99%, AUC 0.98. Capillary (n = 516) 90%, 83%	73%, 53%
Moderate or severe: AUC	0.81 in text, 0.824 in Appendix 8	1.0	Sensitivity 40%, specificity 94%
Severe: sensitivity, specificity, AUC	90%, 80%, AUC 0.93	Venous 82%, 100%, AUC 1.0	31%, 100%

Base paper results, not ours.

Field-representative group (Appendix 6, base paper). Any anemia 66% sensitivity, 69% specificity. Moderate or severe 73%, 69%. Severe 95%, 78%. Accuracy fell in bright outdoor light (data not shown). The authors report that accuracy held across skin tone and phone model.

3.5 The ablation table we are re-examining

This is Appendix 8 of the base paper, measured on the validation set (n = 776). It is the only evidence for the conclusion we are testing.

Inputs	MAE, g/dL	AUC any	AUC mod/sev	AUC severe
Baseline: conjunctiva + metadata	**1.124**	**0.829**	**0.828**	**0.968**
Conjunctiva	1.221	0.819	0.831	0.967
Tongue	1.824	0.656	0.665	0.810
Nails open	1.504	0.749	0.761	0.904
Nails closed	1.575	0.732	0.745	0.875
Conjunctiva + tongue	1.222	0.812	0.821	0.966
Conjunctiva + nails open	1.144	0.832	0.844	0.966
Conjunctiva + nails closed	1.202	0.804	0.831	0.978
Tongue + nails open	1.484	0.760	0.761	0.901
Tongue + nails closed	1.529	0.724	0.745	0.881
Nails open + nails closed	1.446	0.755	0.767	0.907
Conjunctiva + tongue + nails open	1.136	0.825	0.840	0.978
Conjunctiva + tongue + nails closed	1.173	0.804	0.823	0.967
Conjunctiva + nails open + nails closed	1.159	0.813	0.833	0.971
Tongue + nails open + nails closed	1.389	0.765	0.782	0.913
All image sites	1.139	0.822	0.836	0.968
Conjunctiva + metadata + symptoms	1.099	0.837	0.839	0.973
Metadata only: ElasticNet / Random Forest	1.634 / 1.610	0.571 / 0.618	0.611 / 0.625	0.719 / 0.737
Symptoms only: ElasticNet / Random Forest	1.631 / 1.616	0.564 / 0.564	0.608 / 0.608	0.748 / 0.768

Base paper results, validation set, with no confidence intervals reported. On test (n = 777) the baseline scored MAE 1.111 and AUC 0.798, 0.824, 0.927; adding symptoms scored 1.118 and 0.810, 0.819, 0.934.

3.6 Where this table leaves room for a new study

1. **No uncertainty.** It reports point estimates only, with no confidence intervals or tests on 776 people. Many gaps are far too small to interpret by eye.
2. **A possible mismatch in the comparison.** The image rows seem to exclude metadata: conjunctiva alone scores 1.221, and conjunctiva with metadata scores 1.124. If so, "no image combination beat the baseline"

compares images alone with images plus metadata. Among image-only rows, conjunctiva + tongue + nails open (1.136) is lower than conjunctiva alone (1.221). The appendix does not say this outright and the training code is not released, so phrase it as "appears to". Our design removes the issue, since every model gets the same metadata.

3 **The claim is MAE-only.** On AUC, several rows match or exceed the baseline: conjunctiva + nails open has 0.832 and 0.844 against 0.829 and 0.828, and two rows reach 0.978 for severe anemia against 0.968. These may be noise, which is exactly why they need testing.

4 **One split.** Results come from a single validation split with no repeated folds.

5 **One fusion method.** Concatenation is the only one the paper describes.

6 **Preprocessing built for one site.** The paper and its released code describe a crop for the conjunctiva only. How tongue and nails were cropped is not described.

7 **No explanation.** The paper says the result suggests "limited complementary information" but does not test why.

3.7 Inconsistencies inside the base paper

Cite the version you use, and state which one it is. Do not quietly pick one.

Item	Version A	Version B
Learning rate	0.02 (Methods)	0.013 (Appendix 5)
Batch size	64 (Methods)	32 (Appendix 5)
Test MAE	1.12 (main text)	1.111 (Appendix 8)
Moderate/severe test AUC	0.81 (Results)	0.824 (Appendix 8)
Symptoms model, any-anemia AUC	0.937 (Results)	0.837 (Appendix 8)
Ablation MAE unit	mg/dL (Results, clearly a typo)	g/dL (everywhere else)
One-sided significance level	$P < 0.05$ (sample size)	$P < 0.025$ (analysis)
Severe cut-off for ages 5-11	< 8.0 g/dL (Appendix 4, which matches our WHO reading)	< 7 g/dL (released code)
Pregnant share of women	48.1% (Results, whole cohort)	371 of 652, or 56.9% (Appendix 7, test set). Our released-data count is 58.0%.
Block design	Three convolutions per block (Appendix 3 diagram)	Two convolutions per block by default (released code, "basic")
Appendix numbering	Text calls block details Appendix 2 and the schematic Appendix 3	The appendices are the other way round

3.8 Limitations the authors state

- Accuracy remains below invasive point-of-care devices such as HemoCue.
- Hb is over-predicted for severe cases and under-predicted for the normal, mild and moderate groups.
- The hospital population may not represent other skin tones or people with conditions that change these body sites.
- Staff were not blinded to Hb, which affects the clinician comparison.
- Non-consecutive sampling is known to overestimate diagnostic accuracy [45].
- The model needs validation in real community settings.

All of these apply to our study too, because we use the same data. Say so in our Limitations section.

4. Literature review material

How far this has been checked. The descriptions below come from the base paper's own discussion or from our README. We have not read most of these studies in full. Read each one before citing a specific claim from it, and treat the figures as second-hand until then.

Study	Site	What is reported	How to use it
Desai 2026 [1]	Conjunctiva, with tongue, nails and PPG tested	See Section 3.	The base study.
Lin 2025, BPANet [26]	Conjunctiva, palm or nail	One shared ResNet-50 with a body-part tag, taking one image at a time. It does not fuse sites per patient, because no paired dataset existed.	Closest multi-site precedent. Our gap: paired sites per person.
Gowda and Gowda 2026 [27]	Conjunctiva + nail	Late fusion of two CNNs: 82.5% accuracy against 84.5% for conjunctiva alone. No patient-level split or Hb reference.	Weak evidence, useful as motivation only.
Mannino 2018 [14]; Mannino 2025 [15]	Fingernail	2018: app using patient-sourced photos. 2025: about 200,000 US users, MAE ±0.72 g/dL; 89% sensitivity, 93% specificity at 12.5 g/dL; personalisation improved MAE from 1.36 to 0.74. Few moderate or severe cases.	Shows nails can carry a signal, which is why the nail result here needs an explanation.
Collings 2016 [18]; Suner 2021 [16]; Zhao 2024 [17]	Conjunctiva	Digital or smartphone conjunctiva photos. Zhao: 435 emergency department patients, about 75% accuracy.	Earlier conjunctiva evidence, in small or single-setting samples.
Hong 2025 [23]	Conjunctiva, grayscale	Radiomics with machine learning, over 12,000 images from 565 school-age children, without deep learning.	Alternative technical route.
Das 2024 (NiADA) [24]; Sehar 2025 [25]	Conjunctiva	NiADA across 556 people in India: 75.8% and 53.8% in adult women against 23.8% and 90% in pregnant women.	Motivates our subgroup analysis.
Mahmud 2023 [22]	Lip mucosa	138 people, 75-96% accuracy depending on the model. Betel nut or tobacco may limit generalisability.	Other sites, small samples.
Wang 2017 [19]	Fingertip	Unmodified smartphone camera and white flash.	Early camera-based Hb work.
Ramzan 2024 [20, 21]	Conjunctiva + health records	Multimodal fusion and attention mechanisms.	Precedent for attention fusion.
Hasan 2021 [12]; Mohamed 2024 [13]	Reviews	Reviews of phone-based and image-based Hb estimation.	Framing and the variety of methods.
Chalco 2005 [10]; Kalantri 2010 [11]	Clinical pallor	Accuracy of pallor assessment.	Why examination alone falls short.

Themes to build the review around

1 **Feasibility is established but the evidence is uneven.** Studies differ in population, Hb range, reference standard and validation. Direct comparison of headline numbers is unsafe.

2 **Most work studies one site.** The base paper claims to be the only study exploring tongue, conjunctiva and nail beds together, so the question of added value has barely been tested.

3 **Weak validation is common.** Image-level splits inflate performance when one person has several images. This is a reason for our person-level folds.

4 **Subgroups can fail badly.** NiADA shows that a good average can hide poor performance in pregnancy.

5 **Few papers ask why a site fails.** Adding a modality without improvement is usually reported as a final answer. Our diagnosis framework is the new part.

Search suggestions to widen the review

Search PubMed, IEEE Xplore and Google Scholar for: *smartphone hemoglobin estimation deep learning; conjunctiva anemia CNN; fingernail anemia smartphone; tongue image anemia; multimodal fusion redundancy medical imaging; modality contribution Shapley; site-specific preprocessing.* The tongue literature is thin in our current list and needs the most work.

5. Research gap, questions and contribution

5.1 Research gap

A large public dataset now pairs conjunctiva, tongue and nail-bed photos with lab Hb for the same people, but the only analysis of whether these sites complement each other is one table of unadjusted point estimates from a single split. It used one fusion method and has no described site-specific preprocessing. Nobody has established whether tongue and nails are uninformative, redundant, or simply failed by the method.

5.2 Research questions

1 **RQ1.** Which single site or combination gives the best Hb estimation and anemia classification?

2 **RQ2.** When a site does not add value, why? Is it uninformative, truly redundant with the conjunctiva, failed by the model or fusion, or hidden by poor image quality?

3 **RQ3.** Do site-specific models change the answer compared with one conjunctiva-tuned pipeline?

4 **RQ4.** Does the answer differ by subgroup: children or adults, pregnant women, severity, lighting, or phone model?

5.3 Pre-specified main comparisons

These are fixed in `evaluate.py`, within each pipeline and split: **C+T vs C**, **C+N vs C** and **C+T+N vs C**. Every other comparison is secondary and exploratory. State this in the paper, and do not change the list after seeing the results.

5.4 The diagnosis framework

This is the central idea of the paper: for each site that does not add value, decide which explanation holds.

Explanation	Meaning	Evidence that supports it	Status
Uninformative	The site carries little Hb signal.	Single-site model scores no better than a metadata-only model.	Needs the metadata-only baseline (Section 7).
Truly redundant	Signal exists but duplicates the conjunctiva.	Single-site model beats metadata, yet misses the same patients as C (error overlap above chance) and has a low Shapley contribution.	Code ready.
Model or fusion failed	Information exists but this model cannot use it.	Result changes with the specific pipeline or a different fusion method.	Specific pipeline waiting on crop review. Other fusion methods not built.
Bad images	Henna, blur, glare or cold fingers hide the signal.	Result changes when restricted to or retrained on good-quality images.	Subgroup split only. Retraining not built.

Fix the decision thresholds before results are seen, for example what margin counts as "beats metadata". If they are set after, say so in the paper. These explanations can also overlap, so the paper should allow for more than one per site.

5.5 Contributions we can claim, if the work completes

- The first head-to-head comparison of all seven site combinations on this dataset with cross-validation, confidence intervals and paired tests.
- A shared-versus-site-specific preprocessing comparison, which tests whether earlier site rankings depended on the pipeline.
- An explicit framework for explaining why a modality fails to add value, applied per site.
- Subgroup-level answers relevant to deployment.
- Documentation of how the released dataset maps onto the published cohort, including the person-level identifier and the relabelling needed for WHO cut-offs.

Check each "first" claim against the literature before using it. These are planned contributions, and they hold whichever way the result comes out. Confirming the base paper's conclusion on firmer evidence is still a useful finding.

6. Dataset and methods

6.1 Dataset

Aspect	Detail
Source	`sewa-rural-care/anemia-survey-dataset` on Hugging Face, v1.0, 2026 [5]
Access and licence	Gated with approval. CC BY-NC 4.0 plus a data use policy: non-commercial, no re-identification, no redistribution of raw files. The data is not in our repository, and everyone downloads their own copy.
Size	734 Parquet shards, 5,869 rows, about 650 GB in total. Almost all of that is PPG (about 110 MB per row). Without PPG frames, about 5.1 GB.
How we fetch it	`fetch_subset.py` reads only the wanted columns of each remote shard using HTTP range requests, so PPG bytes are never downloaded. Resumable, with a fixed shard order (seed 42).
Columns kept	`patient_uuid`; 4 photos; camera metadata; all `survey_*` (demographics, symptoms, examination, light, device, skin tone, HemoCue); all `cbc_*`; `hemoglobin_category`; PPG metadata.
Photo size	Mainly 640 x 480 in a sample of 60 shards.

6.2 What the raw table does not tell you

We found these by inspecting the data. They are not in the dataset card or the base paper, and they are worth reporting because other users could get them wrong.

- `patient_uuid` **identifies a capture, not a person.** Field captures carry an `_INDOOR_n` or `_OUTDOOR_n` suffix, and 29 people appear 3 or 4 times. The person key is the first 36 characters, giving 5,782 people.
- **The field group can be identified.** It is the 516 rows with a blank `survey_data_collection_place`, from 429 people. This matches the paper's Appendix 6: 429 people plus 3 extra images from each of 29 people gives 516. This match is our inference, since no flag marks the group. Repeat captures have only the conjunctiva photo.
- **7 rows are unusable.** 4 have no lab Hb, 2 have no survey data, and 1 has its survey columns shifted by one.
- **Numeric columns have mixed types.** They are strings in most shards and numbers in 2 shards, which broke the supplied loader until we fixed it.
- `hemoglobin_category` **uses fixed adult cut-offs.** Applying WHO cut-offs changes 1,834 labels: 1,681 are pregnant women and 59 are under 15. Our README originally framed this as a problem mainly for children. It mostly affects pregnancy.
- **The released development data is not the published cohort.** 5,869 rows less 516 leaves 5,353, compared with the paper's 5,455 recruited and 5,176 analysed. There is no flag for the paper's 279 exclusions or its exact split, so our numbers cannot be matched row by row to theirs.

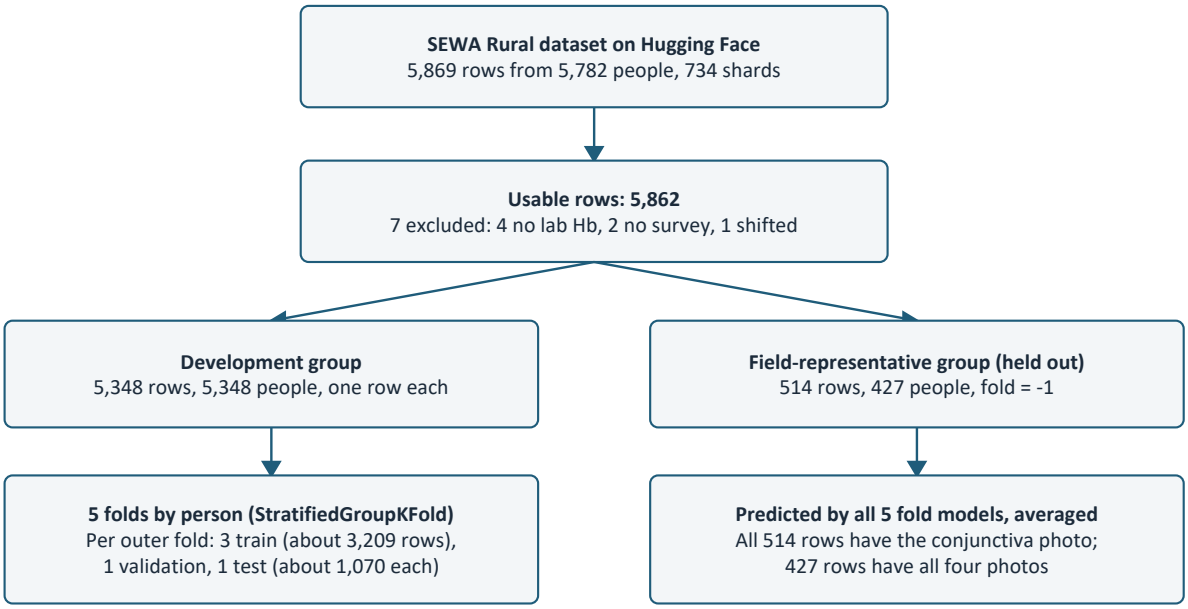


Figure A. Cohort flow as used in our experiments. Redraw it for the paper.

6.3 Cohort description

Characteristic	Development (fold 0-4)	Field-representative
Rows, people	5,348, 5,348	514, 427
Age, years: mean (SD), median (IQR)	24.8 (11.6), 24 (18-29)	27.3 (10.1), 25 (22-30)
Age range	1.1 to 85	5 to 72
Under 15 years	772 (14.4%)	28 (5.4%)
Female, male, other	4,483, 860, 5	477, 37, 0
Pregnant	2,601 (58.0% of women)	304 (63.7% of women)
Hb, g/dL: mean (SD), range	10.05 (2.25), 1.3 to 19.2	10.45 (2.19), 1.4 to 15.1
Normal, mild	1,460 (27.3%), 1,330 (24.9%)	194 (37.7%), 131 (25.5%)
Moderate, severe	1,801 (33.7%), 757 (14.2%)	142 (27.6%), 47 (9.1%)
Phone: A05, A14, S24	1,995, 1,577, 1,776	161, 217, 136
Season: winter, monsoon, summer	2,316, 1,594, 1,438	514, 0, 0
Place	Indoor 5,327, outdoor 18, other 3	Suffix: indoor 302, outdoor 212
Nail polish or henna reported	101 (1.9%)	22 (4.3%)
HemoCue available: venous, capillary	5,348, 3,586	514, 514

Descriptive statistics of our usable cohort, with WHO bands from cbc_hgb_g_dl. Rows, not people, in the field column. Use this as the basis for Table 1.

Severity bands are balanced across folds: each fold has 266 mild, 360-361 moderate, 292 normal and 151-152 severe rows.

6.4 Anemia labels

Population	Normal	Mild	Moderate	Severe
Under 2 years	≥ 10.5	9.5-10.4	7.0-9.4	< 7.0
2-4 years	≥ 11.0	10.0-10.9	7.0-9.9	< 7.0
5-11 years	≥ 11.5	11.0-11.4	8.0-10.9	< 8.0

Population	Normal	Mild	Moderate	Severe
12-14 years	≥ 12.0	11.0-11.9	8.0-10.9	< 8.0
Women 15+, not pregnant	≥ 12.0	11.0-11.9	8.0-10.9	< 8.0
Men 15+	≥ 13.0	11.0-12.9	8.0-10.9	< 8.0
Pregnant (1st/3rd trimester row)	≥ 11.0	10.0-10.9	7.0-9.9	< 7.0

Hb in g/dL, from base paper Appendix 4 and WHO 2024 [6]. Three assumptions of ours: trimester is not recorded, so pregnancy uses the 1st/3rd trimester row; adults over 65 keep the adult row; "Other" sex (5 rows) takes the men's edge, which flags rather than clears. The released code uses the women's edge instead.

The classification tasks are **any anemia** (below the mild edge), **moderate or severe** (below the moderate edge) and **severe** (below the severe edge). Because the edge differs per person, the ranking score for AUC is (cut-off minus predicted Hb).

6.5 Splitting

- `StratifiedGroupKFold`, 5 folds, seed 42, stratified on WHO band and grouped by person [47]. One fixed assignment is used for every experiment. `prepare_cohort.py` is deterministic: two runs gave an identical file hash.
- For outer test fold k, fold k+1 (mod 5) is validation and the other three are training. An assertion in `train.py` checks that no person is in both training and test.
- The field group is never used for training or selection. Each fold's model predicts it, and `evaluate.py` averages the five predictions.
- All models are compared on the rows that every model can use, so field comparisons with tongue or nails use the 427 rows with all four photos.

6.6 Preprocessing

Step	Shared pipeline	Specific pipeline
Idea	The conjunctiva recipe applied to every site unchanged. This is close to the likely default of pushing all sites through one pipeline, and is our control.	A region-of-interest and colour treatment tailored to each site.
Conjunctiva	Find the white paper (L* above 150, near-neutral colour, opening with a 15 px kernel, largest component, at least 5% of the image). Rotate so it is on top. Take the largest centred square directly under its edge.	Identical to shared.
Tongue	Same recipe as the conjunctiva.	Threshold a* at max(Otsu [39], 135), opening with a 9 px kernel, largest component, area 3-60% of the image. Square around it, 1.1 x its largest extent. Fill glare by inpainting [40].
Nails, both views	Same recipe as the conjunctiva.	White-balance so the paper's median colour maps to 235. Treat the largest non-paper region as the hand, and find its main axis by SVD. The end touching the image border less is the fingertips. Square of half the hand length, centred 20% of the way in from the tips. Fill glare.
Glare rule	None.	L* above 235 and near-neutral colour; inpainted with Telea's method [40] (radius 3).
If the crop fails	Centre square.	Centre square, with glare filled.
Then, always	Resize to 256 x 256 by area interpolation, convert to CIELAB, 8-bit.	Same.

Crop found in	Conjunctiva	Tongue	Nails open	Nails closed
Shared pipeline	97.3%	24.8%	36.6%	49.4%
Specific pipeline	97.3%	93.5%	99.8%	99.8%

Share of photos where the rule found its target. This shows the rule fired, not that the crop is correct. Visual review of the specific crops is still pending. Where the shared crop fails, the model sees a centre square of the photo.

Why shared and specific both matter. The shared rule expects a paper edge with the site directly below it, which matches the eye photo. In nail photos the paper sits under the hand instead. So a poor tongue or nail score under this pipeline may reflect the crop rather than the site. The README's execution order exists for this reason: comparing sites only through a conjunctiva-tuned pipeline would bias the ranking before the analysis starts.

6.7 Model and training (as currently running)

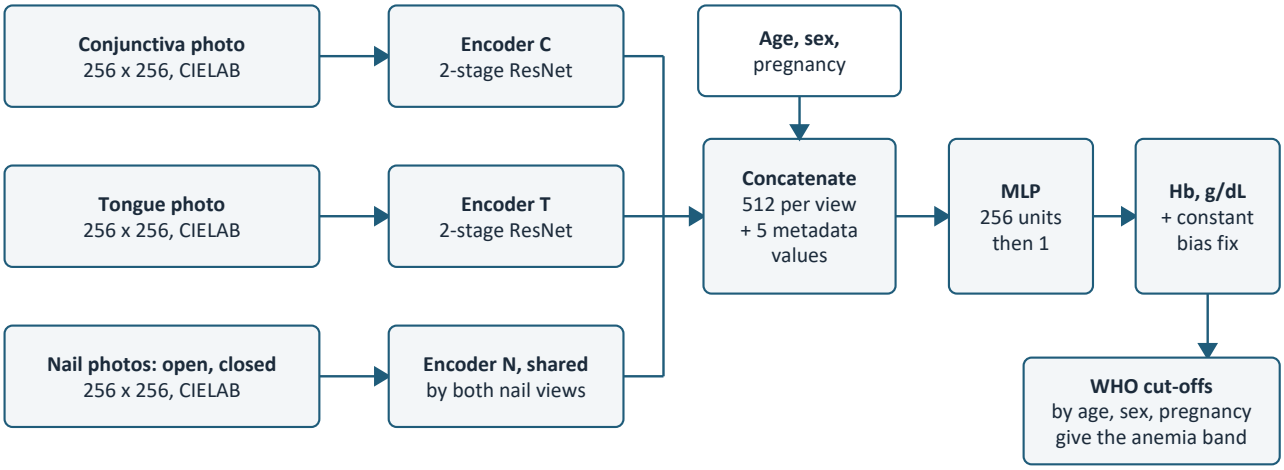


Figure B. The model for C+T+N. Single-site and two-site models use only the matching encoders. Redraw it for the paper.

Component	Setting
Encoder (one per site)	7x7 stem, stride 2, 64 filters, batch norm, ReLU, 3x3 max pool stride 2. Then two residual stages of one block each. Each block is 1x1, 3x3, 1x1 convolutions with batch norm, and a projected skip connection [29]. Stage widths are 64 and 128, expanded 4 times, so stage outputs have 256 and 512 channels. Global average pooling gives 512 features.
Nail views	Open and closed views pass through one shared encoder, and both feature vectors are kept. N therefore has 2 views.
Metadata	5 values: age/100, 3 one-hot sex values (female, male, other), pregnant flag. Every model gets them, which keeps the comparison fair.
Fusion	Concatenate all view features and metadata, then one 256-unit hidden layer with ReLU, then 1 output. The output is added to the training mean Hb.
Input scaling	CIELAB bytes divided by 127.5, minus 1.
Augmentation	Random horizontal flip per image at training time.
Loss	Huber [30], delta 1.0, weighted per sample.
Sample weight	1 g/dL bins, weight proportional to count$^{-0.5}$, normalised to mean 1 over training. The square-root form comes from deep imbalanced regression [31].
Optimiser	Adam [32], learning rate 0.0001, exponential decay 0.97 per epoch. Batch 32, 150 epochs, mixed precision.
Checkpoint	Epoch with the best validation MAE.

Component	Setting
Bias correction	One constant: mean(true Hb minus predicted Hb) on validation, added to all predictions.
Size	596,801 parameters for C. Grows with each added site.
Seeds	Fixed seed 42. `cudnn.benchmark` is on, so runs are not bit-for-bit repeatable, and one seed per run is used.

6.8 Evaluation and statistics

`evaluate.py` pools out-of-fold test predictions, so each development person is predicted once by a model that never saw them. **Compare against our own retrained baseline on our folds**, not the base paper's published numbers, because the splits differ.

Analysis	Definition	Answers
Hb error	MAE, share within ±1 g/dL, bias, limits of agreement (bias ± 1.96 SD) [33].	RQ1
Classification	AUC for the three tasks using (cut-off minus prediction), and sensitivity at 80% specificity.	RQ1
Confidence intervals	Percentile bootstrap, 1,000 resamples [37] of **people, not rows**, since some people have several rows.	All
Paired model differences	A minus B on identical rows: MAE difference, tested by a z-test on per-person differences. AUC difference tested by paired DeLong [34]. Bootstrap intervals for all.	RQ1, RQ2
Multiple comparisons	Holm correction [35], applied separately to main and secondary comparisons within each pipeline and split.	All
Error overlap	For two single-site models: share both wrong, against the product of their error rates (what independence predicts), and Jaccard index.	RQ2
Shapley contribution	For site i among sites S: sum over subsets K of S without i of $\|K\|!\,(\|S\| - \|K\| - 1)! / \|S\|!$ x [v(K + i) - v(K)] [36]. v is -MAE or any-anemia AUC. For the empty set: MAE of predicting the mean, or AUC 0.5.	RQ2
Subgroups (minimum 30 rows)	Under 15, 15 and over; pregnant, non-pregnant women, men; each severity band; each phone; lowest and highest light tertile; clean or marked nails.	RQ2, RQ4
References	HemoCue venous and capillary on the same rows, and clinician examination sensitivity and specificity for any anemia.	Context
Shared vs specific	Compare each combination across pipelines on the same rows. **Not implemented yet** (Section 7).	RQ3

6.9 Our implementation compared with the base paper

> **New finding.** The paper's Data Sharing statement links code at github.com/Google-Health/google-health/tree/master/non-invasive-anemia-ai [3]. It holds preprocessing, an *untrained* single-image model and evaluation helpers, but no training script and no multi-site model. Our README's "not stated" entries were written without it.

Item	Paper and appendix	Released code	Ours, running now
Stem	Not stated	7x7 stride 2, **128** filters, 3x3 max pool	7x7 stride 2, **64** filters, 3x3 max pool
Blocks	[1, 1]; diagram shows 3 convolutions per block	Default "basic": **2** 3x3 convolutions per block	Bottleneck: 1x1, 3x3, 1x1 (3 convolutions)
Stage output channels	Not stated	**128, 256**	**256, 512**
Image features	Not stated	**256**	**512**

Item	Paper and appendix	Released code	Ours, running now
MLP	Not stated	256 + 3 to **64** to 1	512 per view + 5 to **256** to 1
Parameters	Not stated	About 1.25 million (our estimate)	596,801
Metadata	Age, sex, pregnancy	**3** values; encoding not shown	**5** values: age/100, sex one-hot, pregnant
Finding the paper edge	Canny, "heuristic"	Canny (80, 120), then Hough lines (threshold 20)	Bright, neutral colour mask
Crop	Not stated	Fixed x window 120-400 from the edge line to image bottom, rotated to the edge, then stretched to 256 x 256	Largest centred square under the edge, so no stretching
Resize method	Not stated	**Nearest** neighbour	**Area** interpolation
Colour	CIELAB	CIELAB. Decoded image may be BGR when passed to the RGB converter; confirm by reading the code	CIELAB, converted correctly from BGR
Learning rate	0.013 or 0.02, with decay 0.97	No training code	0.0001, with decay 0.97 per epoch
Batch, epochs	32 or 64; 150	No training code	32; 150
Huber delta	Not stated	No training code	1.0
Reweighting	Inverse bin frequency, distribution-based widths	No training code	Inverse square root, 1 g/dL bins
Severe cut-off, ages 5-11	< 8.0	**< 7**	**< 8.0**. This changes 23 children's labels compared with the code
"Other" sex	Not stated	Women's edge	Men's edge
Split	One 70:15:15 random split	Not included	5 folds by person
Exclusions	279 removed, plus pathologist QC	Not included	Only 7 unusable rows; failed crops fall back to a centre square

6.10 Tuning evidence (development runs, not results)

These runs used C on fold 0 and are reported because the settings depart from the paper. They were used for method selection only. Selection was on validation MAE, but we had seen fold 0 test scores while diagnosing the training, so fold 0 is not fully untouched. Disclose that.

Learning rate	Weighting	Epochs	Best validation MAE
0.013 (paper's Appendix 5)	Plain inverse frequency	40	1.50
0.001	Plain inverse frequency	40	1.41
0.001	None	40	1.17
0.0001	None	40	1.15
0.0001	None	150	1.14
0.0001 (chosen)	**Inverse square root**	**150**	**1.16**

Our architecture only. These runs do not show that 0.013 fails on the published architecture, which has not been tried.

Why we chose square-root weighting despite a slightly higher overall MAE. On validation it gave lower error in severe cases (1.61 against 1.77 g/dL) and a higher severe AUC (0.941 against 0.929). Severe anemia is the clinical priority in the base paper, and it uses reweighting too. State this trade-off openly.

6.11 Software, hardware and reproduction

Item	Detail
Hardware	One NVIDIA GeForce RTX 3050 6 GB laptop GPU, 16 logical CPU cores. About 19 minutes per conjunctiva-only model.
Training environment	Python 3.11, PyTorch 2.6.0 with CUDA 12.4, scikit-learn 1.9.0, OpenCV 5.0.0, SciPy 1.17.1, pandas 3.0.5, NumPy 2.4.6.
Download environment	pyarrow 24.0.0, huggingface_hub 1.33.0, datasets 5.1.0, Pillow.

```
python fetch_subset.py --mode images --budget-gb 6      # 734 shards, 5,869 rows
python prepare_cohort.py                                  # cohort.csv with labels and folds
python preprocess_images.py --out C:/tmp/ds-research/images
python preview_crops.py --images C:/tmp/ds-research/images  # contact sheets: never share them
python run_experiments.py --images C:/tmp/ds-research/images  # resumable
python evaluate.py                                        # writes data/results/*.csv
```

File	Purpose
fetch_subset.py, load_subset.py	Download selected columns and load them as a dataset.
prepare_cohort.py	Person IDs, exclusions, WHO labels, field group, folds.
preprocess_images.py, preview_crops.py	Both pipelines, crop status, contact sheets.
train.py, run_experiments.py	One run; the whole experiment grid.
evaluate.py	Overall, comparisons, error overlap, Shapley, subgroups, references.
data/runs/<pipeline>/<subset>/fold<k>/	predictions.csv, summary.json, best.pt (not in git).
data/results/	The CSVs that feed the paper's tables (not in git).

7. What is not done, and decisions still open

Status on 1 November 2026. Whoever writes the paper should check this list again, because Methods must describe only what actually ran.

Item	Why it matters	Current state	To finish
Model differs from released code	The README calls C a "reimplementation". Ours differs in width, block type, MLP and crop. Any large architecture difference could change the C baseline.	Running with our assumed model. Discovered 1 Nov.	**Team decision.** Test the released architecture on C using validation data. If it clearly beats ours, restart. If it doesn't, keep ours and report it as an adaptation.
Metadata-only baseline	Every model gets metadata. A weak image site could score the same as metadata alone, and only this baseline can show it. The "uninformative" test needs it.	Not built.	Train on the same folds with no image. Fast, and needs little or no GPU.
Specific pipeline	RQ3 and the "model failed" test depend on it.	Code ready. About 24 hours, not started.	Someone must look at the tongue and nail contact sheets, then run it.
Per-site hyperparameter search	The README requires the same search for every site. The learning rate was chosen on C only, which could favour C.	Not done.	Run the same small search on validation for T and N.
Other fusion methods	Needed for "model or fusion failed".	Only concatenation exists.	Add late averaging (cheap, from existing predictions) and attention (needs training) [41, 42].
Good-quality retraining	Needed for "bad images". Currently only a subgroup split at evaluation.	Not built.	Define quality rules, then retrain on them. Blur and glare measures are available beyond the survey flags.
Shared vs specific test	RQ3 needs a formal paired comparison.	`evaluate.py` only compares within a pipeline.	Add it, with Holm correction.
Sensitivity and specificity at WHO cut-offs	The paper's headline metric and the 80% target. We only compute sensitivity at 80% specificity.	Not computed.	Add to `metrics()`.
Holm on DeLong p-values	Correction is only applied to the MAE p-value.	Partial.	Extend to AUC tests.
Skin tone subgroup, ICC, R^2	The paper reports them, and skin tone fairness matters here.	Not computed.	Add ITA° from `survey_skin_tone_l/b`. Check the paper's formula: standard ITA is arctan((L*-50)/b*) x 180/π [46].
Plots	Bland-Altman, ROC and scatter plots are needed.	None yet.	Script them from `predictions.csv`.
Repeat seeds	Same settings gave a slightly different result between runs, so tiny gaps are noise.	One seed per run.	Report this. Repeat the key models if time allows.
Code commit	Reproducibility claims need a public, tagged commit.	Nothing committed.	Commit code only. Never `data/`.

> **Allowed wording while these are open.** "We evaluated concatenation fusion", not "we evaluated several fusion methods". "Tuned on the conjunctiva and applied to all sites", not "tuned per site". "Adapted from" the base paper's model, not "reimplemented", unless the architecture decision changes that.

8. Research paper format

The structure below suits a diagnostic-accuracy and machine-learning paper. Follow **STARD 2015** [43] and **TRIPOD+AI** [44] checklists, and adjust to the target venue's template and word limit. Suggested lengths assume a journal paper of about 4,000-5,000 words.

Title options

- Do tongue and nail-bed images add value beyond the conjunctiva? A cross-validated re-examination of smartphone anemia screening
- Site-specific deep learning for smartphone anemia screening: re-examining modality contribution and redundancy
- Redundant or under-modelled? Testing tongue and nail-bed images for smartphone hemoglobin estimation

Choose one once the results are known. Don't let a title promise a finding that the data doesn't support.

Section	What it must discuss	Source here	Length
Abstract	Background; objective; data (5,348 development and field group); methods (7 combinations, 2 pipelines, 5 folds); main results with intervals; conclusion. Write it last.	Sections 1, 5, 6	250-300 words
Keywords	Anemia; hemoglobin; smartphone; conjunctiva; multimodal deep learning; modality contribution.		5-6 terms
1. Introduction	Burden of anemia and why detection falls short. Promise of smartphones. Base paper's result and why it is not settled. The gap in one paragraph. Research questions and contributions, as a list.	Sections 2, 3.6, 5	500-700 words
2. Related work	Smartphone Hb estimation by site. Multi-site work. Multimodal fusion and modality contribution. Validation weaknesses. End by stating what is missing. Use a summary table.	Section 4	500-800 words
3. Materials: dataset	Source, ethics of the original study, licence and our approval. Capture and ground truth. Cohort flow (Figure 1), exclusions, person-level ID, field group, WHO relabelling. Table 1.	Sections 6.1-6.4	400-500 words
4. Methods	Splitting. Both preprocessing pipelines with thresholds. Model and training, with a table of every departure from the base paper. Experiment grid. Fusion methods. Diagnosis framework with rules set in advance. Statistics, including the main comparisons.	Sections 5.3-5.4, 6.5-6.11, 7	1,000-1,400 words
5. Results	Findings only, without interpretation, ordered by research question. See the planned tables and figures below.	`data/results/`	800-1,000 words
6. Discussion	Principal findings. Answer per site: which explanation holds, and how confident we are. Agreement or disagreement with the base paper and possible reasons. Clinical meaning against the 80% target and HemoCue. Deployment implications: which photos are worth capturing, and who benefits.	Sections 2.3-2.4, 3.8	800-1,000 words
7. Limitations	See the list below. May sit within Discussion.	Below	200-300 words
8. Conclusion	Direct answers to the research questions in plain terms. Future work. No new claims.		100-150 words
Declarations	Data availability (gated, link to the source; we do not redistribute). Code availability. Ethics (secondary analysis; check local requirements). Funding. Competing interests. Author contributions. Acknowledge SEWA Rural and the participants.	Section 6.1	Short
References	In the venue's style.	Section 10	

Section	What it must discuss	Source here	Length
Supplement	Full subgroup tables, full pairwise comparison table, crop rules, cut-off table, tuning runs, STARD and TRIPOD+AI checklists.		As needed

Planned tables

#	Table	Contents	From
1	Cohort	Development and field characteristics.	Section 6.3
2	Departures from the base paper	Item-by-item comparison.	Section 6.9
3	Overall performance	7 combinations x 2 pipelines: MAE, within ±1, 3 AUCs, sensitivity and specificity, all with 95% CIs. One table for test, one for field.	overall.csv
4	Main comparisons	C+T, C+N, C+T+N vs C: differences, CIs, Holm-adjusted p.	comparisons.csv
5	Shared vs specific	Paired differences per combination.	To add
6	Diagnosis summary	One row per site: verdict and supporting numbers.	All
7	Subgroups	Best model vs C by subgroup.	subgroups.csv
8	References	Models vs HemoCue and clinician examination on the same rows.	references.csv

Planned figures

#	Figure	Purpose
1	Cohort flow diagram	STARD requirement. Based on Figure A.
2	Pipeline and architecture	Based on Figure B. Use a drawing, not patient photos, unless the data policy clearly allows otherwise.
3	MAE by combination with error bars, both pipelines	The headline figure.
4	ROC curves for the 3 tasks	Best model, C, and HemoCue and clinician operating points.
5	Bland-Altman for the best model and C	Agreement and severity-dependent bias.
6	Error overlap heatmap and Shapley bar chart	The redundancy evidence.
7	Subgroup forest plot	RQ4 at a glance.

Abstract skeleton

Background: Smartphone photos could screen for anemia without blood sampling. A recent large study reported that tongue and nail-bed images did not improve on the conjunctiva, but gave no confidence intervals for that comparison and did not test why. **Objective:** To determine whether tongue and nail-bed images add value beyond the conjunctiva and, if not, why. **Methods:** Using the public SEWA Rural dataset (5,348 development participants; a further field-representative group of 427 people and 514 images) with venous hemoglobin as reference, we trained all seven site combinations using 5-fold person-level cross-validation under shared and site-specific preprocessing. We compared them with bootstrap confidence intervals and paired tests, and examined error overlap and Shapley contributions. **Results:** [RESULT PENDING: best combination, MAE with 95% CI, AUCs, main paired differences, field results, diagnosis per site, subgroup differences]. **Conclusion:** [RESULT PENDING: direct answer, and what it means for field capture].

Limitations to include

- Same single-hospital, rural Gujarat population as the base paper; not a community sample. External validity is unknown.

- The field-representative group was collected on the hospital campus by the same staff. It is not a truly independent external test.
- Enriched, non-consecutive sampling can overestimate accuracy [45], and our AUCs depend on this case mix.
- The base paper's exact split and exclusions could not be reproduced, so direct numeric comparison is not valid.
- Rows with poor images were not all removed, unlike the published pathologist QC, so our baseline may be lower for that reason.
- Trimester is unrecorded, so pregnancy cut-offs are approximate. Clinician assessments were not blinded.
- Crop rules for tongue and nails are heuristic and were designed by us. A learned segmenter might do better.
- PPG is excluded.
- Small-sample subgroups have wide intervals. Subgroup findings are exploratory.
- Any item from Section 7 that is still unfinished at submission must be listed here.

Future work to mention

- Add PPG, which we excluded for size.
- Learned segmentation for tongue and nails.
- Prospective validation in real community settings, with low-end phones and frontline workers.
- On-device models for offline use.
- A cascaded design: phone screening followed by a point-of-care blood test, which the base paper also suggests.
- External validation in other regions and skin tones.

Writing rules for this paper

- Separate estimation from classification. A lower MAE does not automatically mean better screening.
- A non-significant difference is not proof of no difference. Report the interval and what it rules out.
- Don't compare our numbers directly with published ones as if a higher score means a better method.
- Do not call the field group "external validation" without the caveat above.
- Be careful with "first" claims, and never say "diagnose": this is a screening study.
- Don't use any patient image in the paper unless the data use policy clearly allows it.
- Credit the base paper fairly. We build on their dataset and design, and our work exists because they released it.

9. Presentation outline

This follows the PPT contents you listed. Results slides stay as placeholders until the data is in.

Slide	What to put on it	Source
Title slide	Title, team names, guide, institution, date. Add one line: the question we answer.	Section 1
Introduction	Burden: over 2 billion people, high prevalence in Indian children and women. Why testing is hard. Promise of a phone camera. The question: do tongue and nails add anything beyond the eye? Show the 4 research questions.	Sections 2, 5.2

Slide	What to put on it	Source
Literature review	Base paper summary, with its conclusion and its Appendix 8 highlighted. A compact table of other studies by site. The gap: no CIs, one split, one fusion, no explanation. Two or three slides at most.	Sections 3, 4, 5.1
Methodology: dataset	SEWA Rural: 5,869 rows, 3 phones, 4 photos per person, venous Hb. Cohort flow. WHO relabelling (1,834 rows). Person-level folds and field group.	Sections 6.1-6.5
Methodology: models	7 combinations x 2 pipelines x 5 folds. Shared vs specific crops, shown with a schematic rather than patient photos.	Sections 6.6, 1
Model explanation	Architecture diagram (Figure B). One encoder per site, concatenated with age, sex and pregnancy. Regression to Hb, then WHO thresholds. Huber loss and reweighting in one line each. Diagnosis framework table.	Sections 6.7, 5.4
Results and evaluation	First say how we evaluate: person-level bootstrap, DeLong, Holm. Then [RESULT PENDING]: MAE chart, ROC, main comparisons, diagnosis per site, subgroups.	Sections 6.8, 8
Conclusion	Direct answer per research question. Limitations: one hospital, enriched sampling, heuristic crops. Future work: PPG, segmentation, community validation.	Section 8
References	Only those cited on slides: at minimum [1, 3, 5, 6, 29, 31, 34].	Section 10

If time is short, focus on these points: the base paper's claim, why it isn't settled, the 7 x 2 x 5 design, and the diagnosis framework.

10. References

How to read this list. [1]-[3] and [5] were read directly while preparing this document. [7]-[25], [28], [30], [31], [45] and [46] are taken from the base paper's reference list, and [26], [27] and [4] from our README. The methods references [29], [32]-[44] and [47] are standard works, written from memory unless already in the base paper. **Check all details before submission.**

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